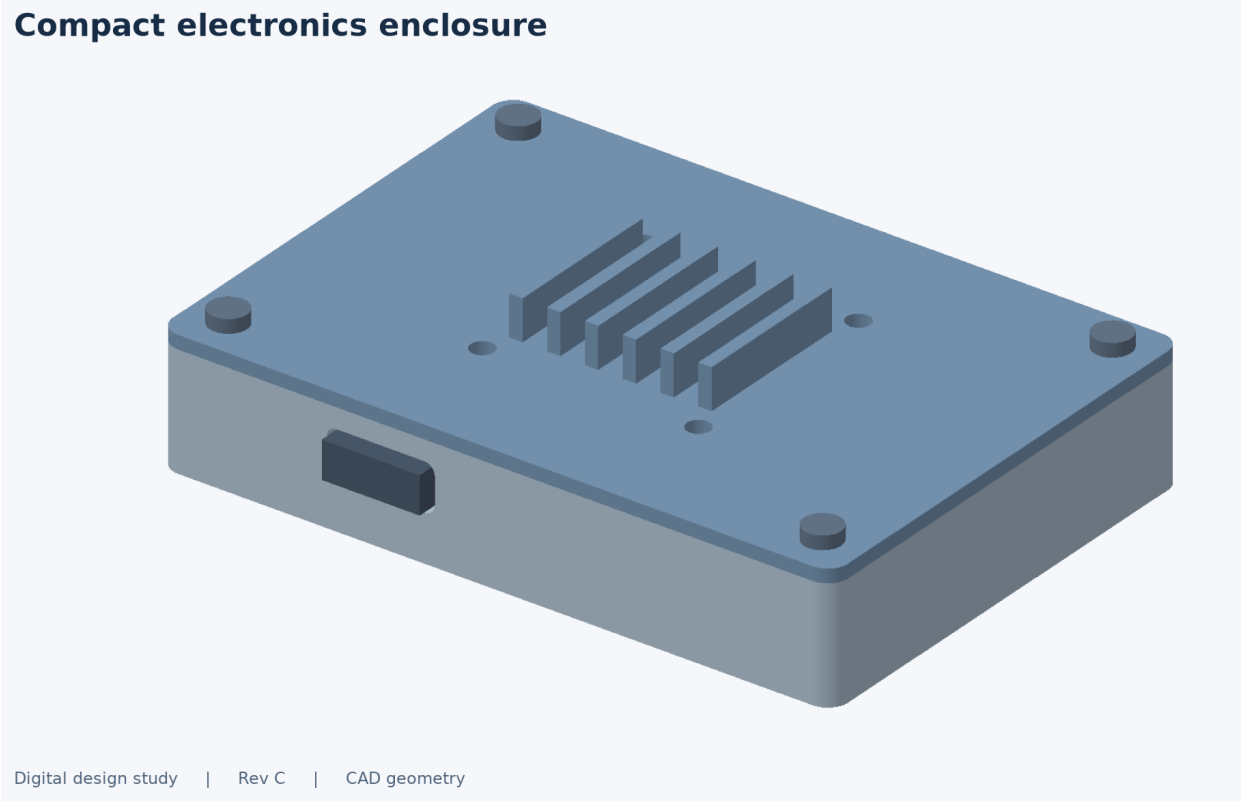
MECHANICAL DESIGN STUDY

Compact edge-compute enclosure

Packaging, thermal contact and manufacturing review



Envelope	Thermal path	Evidence
100 × 70 × 25 mm housing	Chip / pad / integral Al lid / air	PASS nominal digital checks

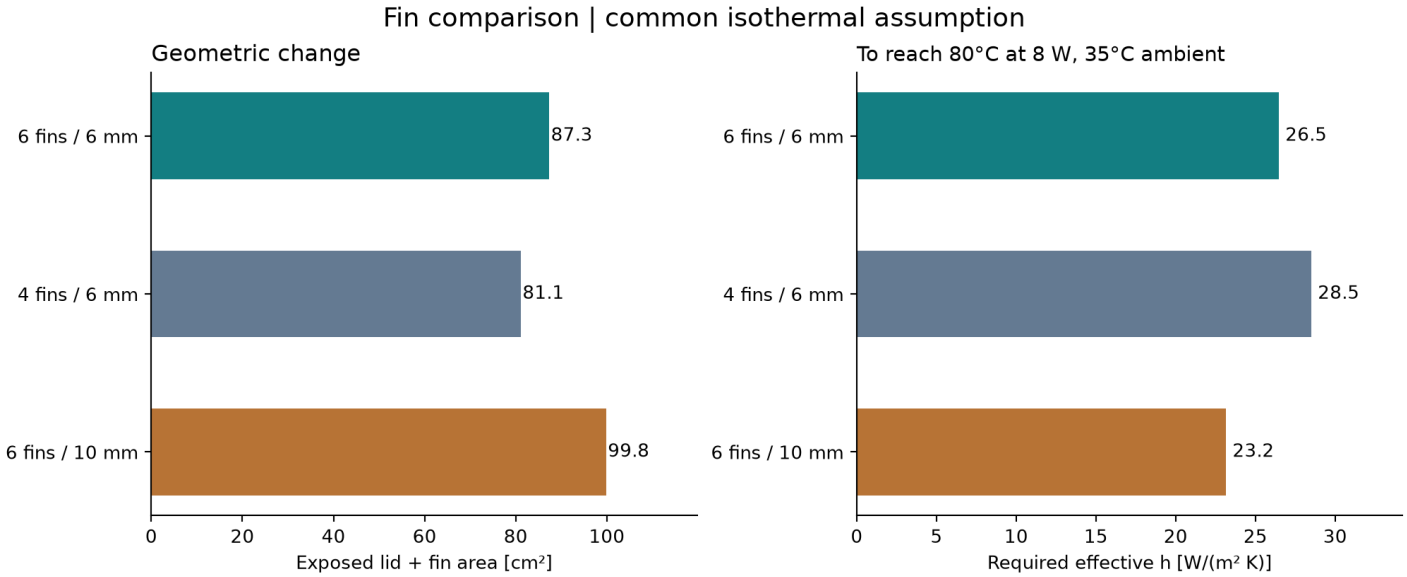
A small enclosure for an assumed electronics board. The removable aluminum lid also acts as its heat sink. This report explains the geometry, the manufacturing approach, the calculations and the decisions that remain unresolved.

Independent demonstrator. Not a comma.ai design or replica. No physical prototype, machining, measured thermal performance, CFD or FEA is claimed.

01 / DIGITAL DESIGN COMPARISON

Fin alternatives and the size constraint

The tray and board stay fixed. Three fin candidates are rebuilt and checked for nominal interference; a fourth is rejected by the machining constraint. This replaces a purely subjective comparison with an executed digital experiment.



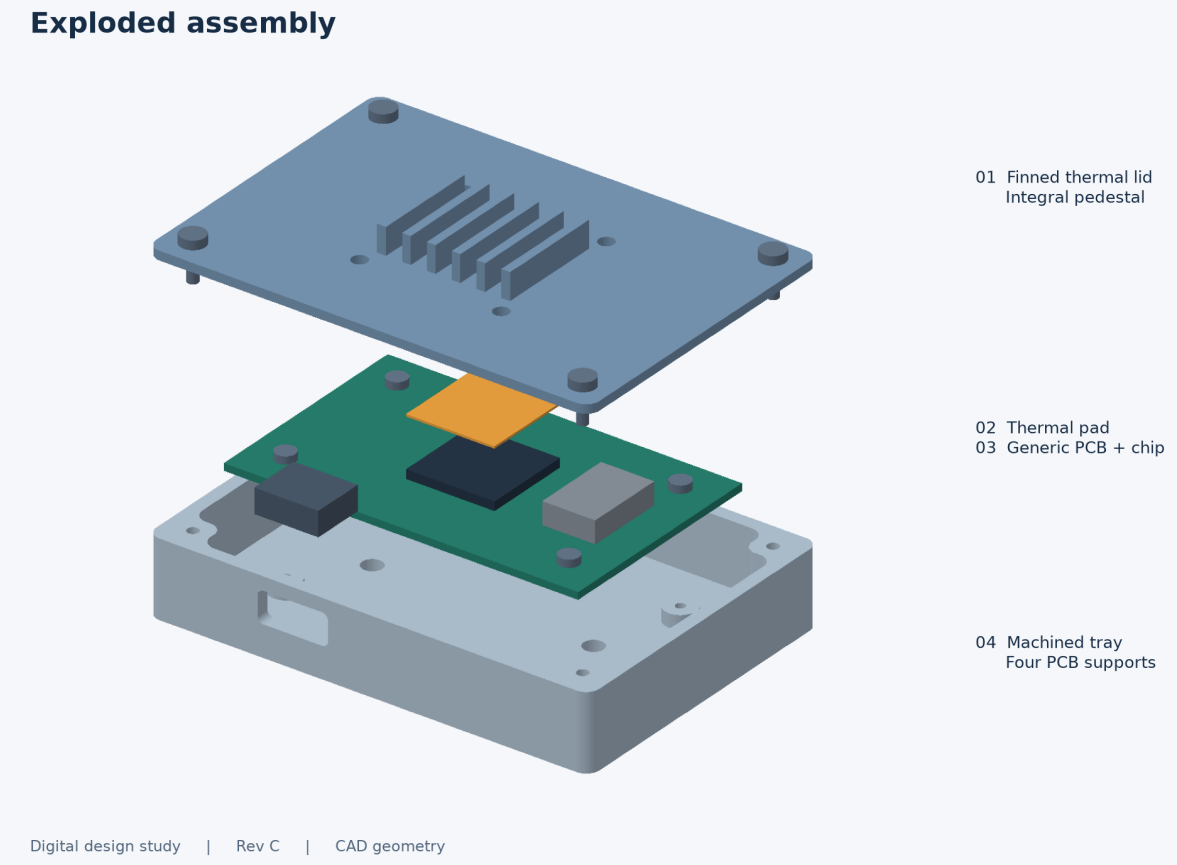
Candidate	Height / area	Engineering consequence
six fins 6mm	25 mm / 87.33 cm²	Within height target; nominal CAD fit checked
four fins 6mm	25 mm / 81.09 cm²	Within height target; nominal CAD fit checked
six fins 10mm	29 mm / 99.81 cm²	Exceeds 25 mm target
Eight fins, 6 mm	Not built	2 mm channels violate the chosen 3 mm cutter constraint.

More area is useful only if it can exchange heat. The area comparison assumes an isothermal lid and uniform effective convection. It excludes radiation and parallel tray/PCB paths. Wider channels can change actual airflow, which this model does not calculate.

Retain the six-fin, 25 mm baseline. Taller fins add area but violate the size target; eight fins require revisiting tooling. Raw inputs, results and derivation: analysis/design-study.md and results/design-study.json.

02 / PACKAGING

The four layers of the assembly



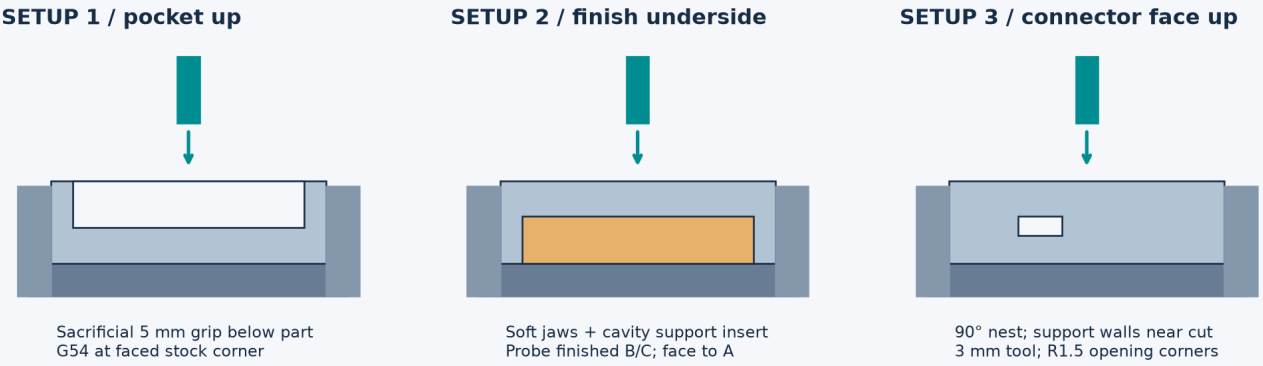
Layer / interface	Nominal geometry
Tray	3 mm floor/wall; 17 mm rim; corner islands and integral PCB supports
PCB and keepouts	80 × 50 × 1.6 at Z6; generic 20 × 20 × 2 chip; separate connector/component envelopes
Thermal lid	2 mm plate at Z17..19; pedestal to Z10.1; six fins to Z25
Hardware	Simplified M3 × 8 lid / M2.5 × 5 PCB screws; actual threads, fan wiring and nuts not fully modeled

The integral pedestal removes loose-heatsink retention ambiguity. Removing the lid separates the thermal interface, so every service cycle requires inspecting and, where the supplier requires it, replacing the pad. All PCB dimensions must be replaced with measured interfaces before ordering metal.

03 / MANUFACTURING

Proposed machining sequence

CNC setup concepts | review before programming



Illustrative setup geometry. No generated or verified machine toolpaths; clamp and tool envelopes require CAM review.

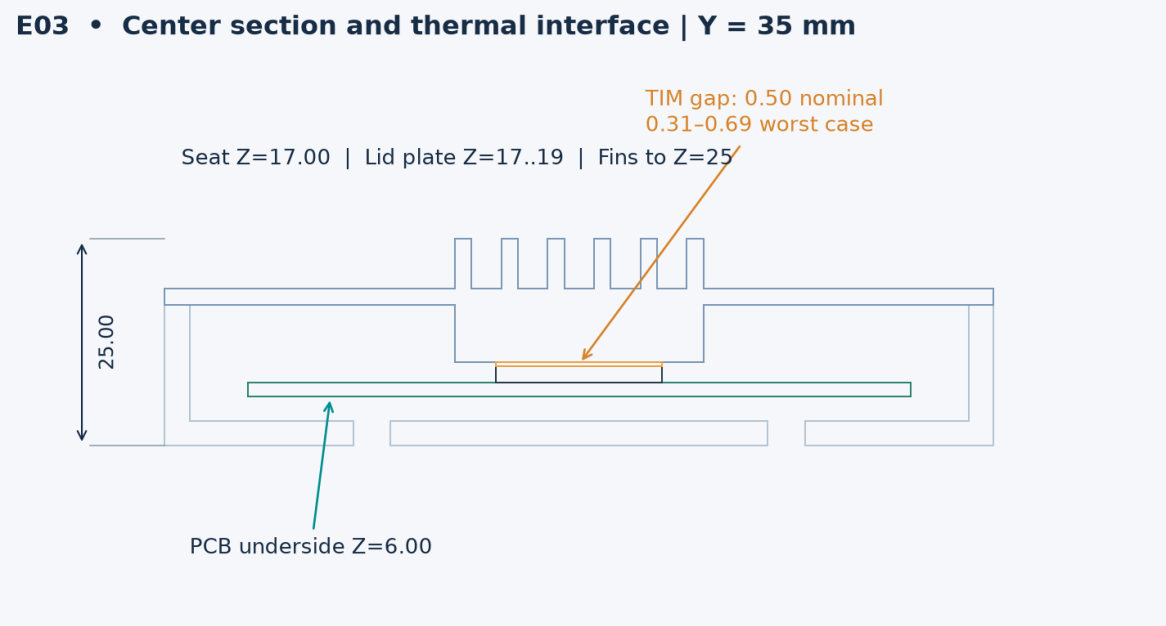
Workpiece	Stock / operation / hold
Tray	110 × 80 × 25 candidate blank. Grip sacrificial foot, pocket from top, flip into supported soft jaws, then rotate for connector opening.
Thermal lid	106 × 76 × 20 blank. Machine pedestal side, flip into relieved support nest, finish 2 mm plate and 6 mm fins. Protect against bow.
Tools	Candidate Ø6 rougher, Ø3 fin/connector finisher, spot/drill/tap tools. Verify reach and holder geometry in actual CAM.
Print precursor	PETG fit print, underside down. Front opening may need bridge support. Printed lid is an unpowered dummy, not a heat sink.

Haas VF-2 is the requested reference machine. The actual configuration, postprocessor, offsets, tool assemblies, clamps and safe simulation remain open. No G-code or production-ready toolpaths are supplied.

Rev B replaced cylindrical corner-boss junctions with a rounded stepped cavity and moved support holes inward. This improves nominal tool access; full swept-tool/holder collision verification is still required.

04 / INSPECTION & STACK-UP

The thermal-pad gap



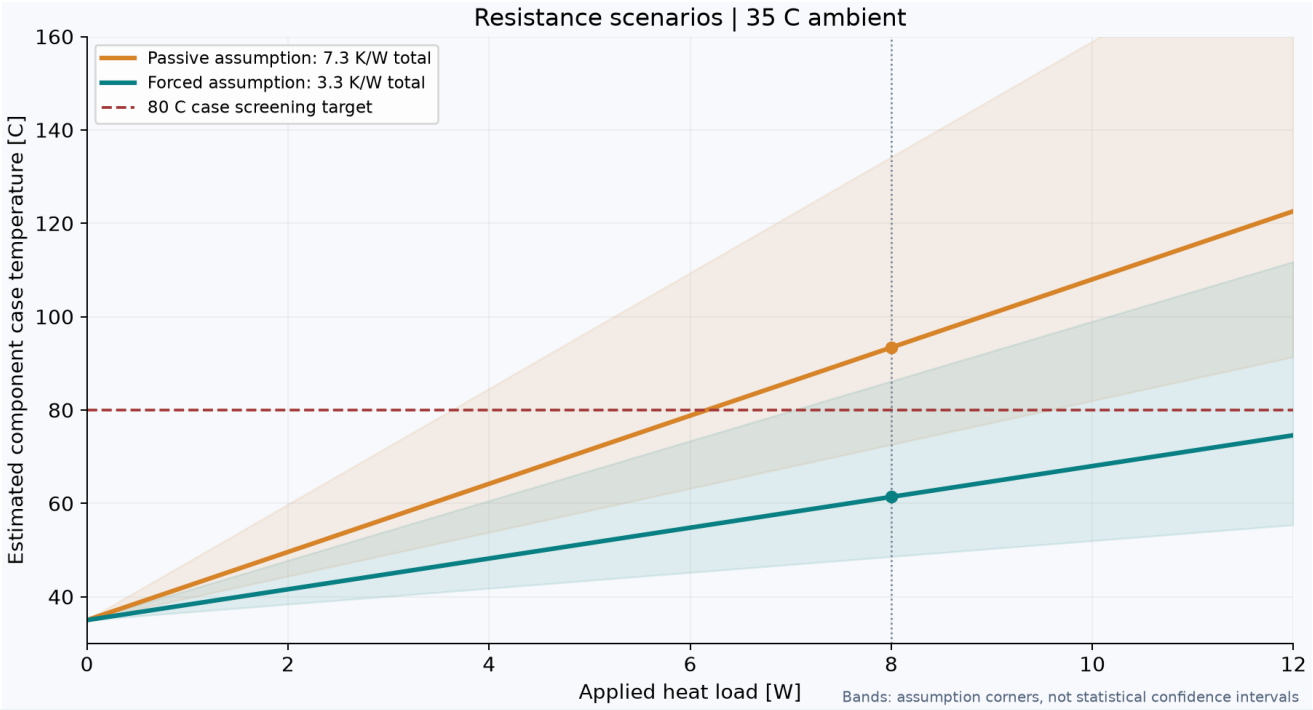
Stack term	Nominal [mm]	Tolerance [mm]
Base rim from A	17.00	±0.03
Minus pedestal projection	6.90	±0.03
Minus PCB support height	6.00	±0.03
Minus PCB thickness / chip height	1.60 / 2.00	±0.05 / ±0.05
Resulting gap	0.50	Worst case 0.31..0.69

A free pad 1.00±0.05 mm would see 27.4..70.5% compression. This is not a released specification: select a supplier pressure curve, measure the real stack and revise the pedestal or shims before loading the board.

A/B/C locate the base. Proposed support height ±0.03 and flatness 0.05 control seating. Hole position uses an actual cylindrical zone and CMM/qualified mandrel inspection; caliper coordinates alone do not certify it. Full dimensions and acceptance methods are in the drawing package.

05 / THERMAL ENGINEERING

What the thermal calculation does and does not show



At 8 W /35°C ambient	Passive	Forced hypothesis
Total thermal resistance	7.3 K/W	3.3 K/W
Predicted case temperature	93.4°C / exceeds 80°C	61.4°C / nominally below 80°C
Uncertainty corner range	72.6..134.2°C	48.6..86.2°C

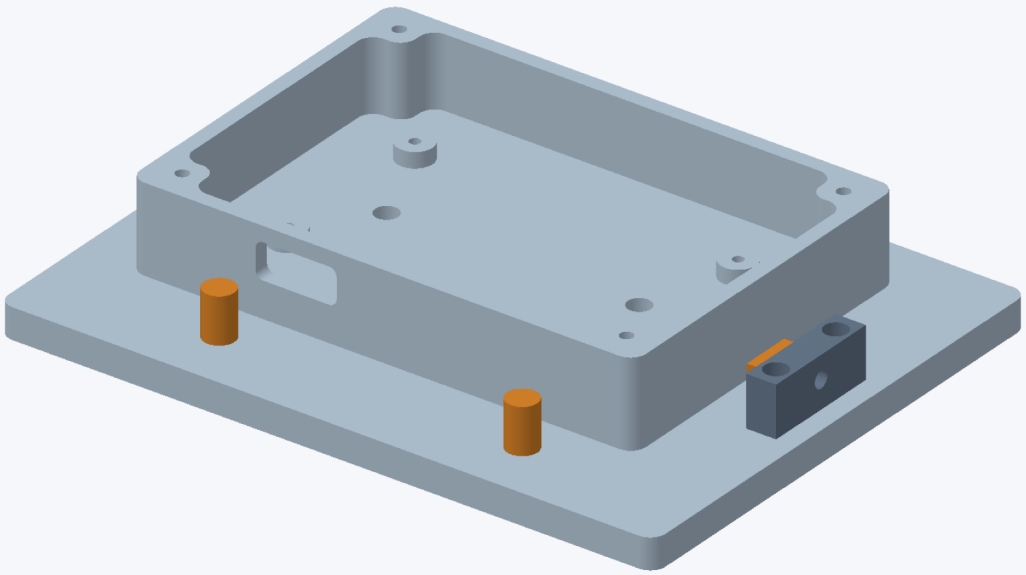
$\Delta T = Q \times R\theta$. The network assumes 0.8 K/W pad/contact,0.5 K/W spreading, and 6.0 or 2.0 K/W effective air resistance. It estimates case temperature, not junction temperature. Bands are assumption corners, not confidence intervals.

The baseline exposed area is 87.33 cm². Its original 6 K/W air resistance implies an effective h of 19.1 W/(m² K); reaching the screening target through the modeled lid path needs about 26.5 W/(m² K). Neither value is established for this geometry. Full derivation: analysis/design-study.md.

06 / FIXTURE & VALIDATION

The separate inspection fixture

Tray in the inspection fixture



Digital design study | Rev C | CAD geometry

Constraint	Implementation
Z / roll / pitch	Three noncollinear support buttons on datum A
Y / yaw	Two front side buttons touching datum B
X	One left button touching datum C
Bias / retention	Two gentle lateral screw pads near the floor; gravity seating. Not a machining clamp.

Use ten complete remove-and-reinstall trials per operator, two operators, with indicators left at the original zero. Proposed per-operator and pooled range ≤0.05 mm; between-operator mean shift ≤0.03 mm. Raw templates intentionally contain no fabricated data.

Replaceable locator buttons can be shimmed; block mounting bores and lateral screw pilots are modeled. Selected screws, nuts, bench feet and a torque-resisting hold-down remain completion items. Do not use this inspection jig for cutting forces.

07 / DIGITAL REVIEW

Scope, evidence and how to review the work

31 component export pairs checked; 38 tests executed, 0 failures; 361 assembly pair checks.

Evidence	Meaning
CAD and exports	Nominal shapes are valid, exported solids reimport, meshes close, and modeled assemblies have no unintended overlap.
Executed design study	Three fin variants rebuilt; eight-fin alternative rejected for cutter clearance. Area comparison uses a stated simplified model.
Tolerances and temperature	Calculated from declared dimensions and assumptions. No physical temperature or pad-pressure measurement is implied.
Manufacturing and fixture	Specific review proposals. No CAM simulation, machine prove-out or repeatability result.

What changed in this review

The obsolete boss-radius parameter was replaced by a real corner-island dimension. Lid-pilot material is now checked against the actual solid. The pad gap follows CAD inputs. Report generation refuses unreviewed input changes. A short reading guide explains the design before exposing the generated files.

Digital scope

The current deliverable is a digital mechanical-design study. Physical fabrication is outside its scope. Further digital work can adapt measured or published board interfaces, use selected pad/fan data, or add machine-specific CAM when a real setup is specified. Cooling, pad force and manufacturing capability remain open engineering questions.

Read and reproduce

Start with START_HERE.md, then analysis/design-study.md. CAD inputs: cad/parameters.json. Calculation assumptions: analysis/assumptions.json. Run **python scripts/rebuild.py**. AI assistance was used to develop the source and documentation; the assumptions and decisions are exposed for owner review.

Public context and sources: report/sources.md. No comma.ai geometry was imported. This is not an official product design.